

President authority and social privilege: A survey experiment among voters in Maharashtra, India

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1 Overview

Not all politicians are equally able to influence decision-making processes and policy outcomes (Tolentino and Cruz 2022; Purohit 2021; Heinze, Brulé and Chauchard 2024). When dominant groups with various sources of social privilege capture political institutions through elected office, this may lead them to accrue an authority advantage. This survey experiment builds on existing literatures on elite capture and political inequality by focusing on two factors that may affect leaders’ authority: leader gender and social class (here: caste).

I employ a survey experiment with a 2x2 factorial design that randomly varies the gender and social class of hypothetical local politicians in order to probe their causal impact on political authority, pliability, and backlash in a face-to-face survey of 2340 voters in Maharashtra, India. This pre-analysis plan was written after project implementation, but before data was received by the author.¹

2 Pilot results

The experimental stimuli and outcome questions were piloted from 12-13 July 2024 in Jalgaon district among a gender- and caste-balanced sample of 10 citizens, 5 living in the Jalgaon City Assembly constituency and 5 in Jalgaon Rural Assembly Constituency, all familiar with rural politics. It was also piloted on 22 July 2024 in the Jalgaon Rural Constituency among 4 voters.

Piloting served three purposes: manipulation checks of the gender and caste variables, construct validity of the outcome variables, and development of options for outcome variables. Table 1 summarizes how the results of the pilot study have informed the design of the main study.

¹The de-identified survey data will be received by the author from Pratik Mahajan upon pre-registration.

Table 1: Pilot study results

Question	Pilot result	Design decision
Do the gendered first names (Bharti and Rohit) lead respondents to correctly identify the gender of the politician?	Yes, all respondents correctly identify Bharti as a female politician and Rohit as a male politician.	Retain these gendered first names for the gender treatment. Remove gender manipulation check after vignette due to space constraints.
Do the caste-specific surnames (Kamble and Marathe) lead respondents to correctly identify the caste of the politician?	Compared to other names like Patil, Marathe leads respondents to universally identify the politician as Maratha. However, some pushback against the idea of the "dominant" or "subordinate" caste in a village in the absence of information about population proportions.	Retain Marathe as the dominant caste surname, and add numerical dominance of the Maratha group as a fixed condition in the vignette, so that it is clear that Kamble would be subordinate and Marathe would be dominant. Remove caste manipulation check after vignette due to space constraints.
What open-ended responses do respondents give in response to outcome questions?	It is possible to categorize follow-up responses into 4-5 categories per sub-question.	Developed sub-category questions based on these 4-5 buckets per sub-question. Allow participants to "select all" where relevant.

3 Design

3.1 Study population, sampling, survey design

2340 voters above the age of 18 were surveyed across 18 state assemblies in North Maharashtra. The survey used a stratified random sampling strategy for surveying voters. 100 voters from each of the 18 constituencies, and then an additional 340 voters in the 3 urban centers were sampled to ensure a balance of rural and urban voters; additional Dalit voters in the Bhusawal constituency and other constituencies were sampled, for a total targeted sample of 2340 voters. The exact gender and ethnic stratification of voters across the 18 constituencies are included in Appendix 1. Within each constituency,

one village or town was selected per 20 respondents, for a total of 4 to 6 sites per constituency, while for the urban centers, all the respondents are sampled in the same city. These sites are selected based on their ethnic composition and areas where grassroots organizations are active in voter mobilization. To achieve the stratified quota sample, a random walk method was employed where enumerators turn left when entering the area and survey every fourth individual until the desired gender and ethnic quota within the location is met. No payment was provided for participation in the survey. The survey took place from late August 2024 to mid October 2024.

The survey included several experimental modules after pre-treatment covariates. The order of the experimental modules – and thus of the experimental module of interest in this study – was randomized.

3.2 Treatment groups and randomization

The experiment features a vignette of a hypothetical village president, where the gender and caste of the president vary, with two possibilities each (2x2 factorial design). Participants are randomly assigned using simple randomization (with equal probabilities of assignment to condition) to 1 of 4 conditions, displayed in Table 2.

Table 2: Experimental conditions

Condition	Gender	Caste
1	Female	Maratha (Dominant)
2	Male	Maratha (Dominant)
3	Female	Dalit (Non-dominant)
4	Male	Dalit (Non-dominant)

In particular, gender of the president was manipulated using she/her or he/him pronouns combined with an obviously female (Bharti) or male (Rohit) first name in this context. Caste of the president was manipulated using surnames with obvious caste connotations in this context: Marathe (Maratha) or Kamble (Dalit). Three other dimensions are kept constant: the age of the president (the mean age of the president

sample from the previous study), the fact that the village is numerically dominated by Marathas, and the amount of time for which the president has been in office (1 year). The vignette wording (in English: the surveyed version is in Marathi or Hindi, depending on the respondent’s preferred language) is as follows, with words in brackets varying according to the experimental condition.²

Now, I’ll tell you about the president of a village council here in Maharashtra. I’m interested in your opinions about this president. [Bharti/Rohit] [Marathe/Kamble] is a 42-year old [female/male] president. [She/he] is a [Maratha/Dalit] president, in a village where more than half of people are Maratha. [She/he] has been the president of the village council for one year.

Based on your experience and knowledge of the way village councils work in reality on the ground, what type of experience do you think [Bharti/Rohit] [Marathe/Kamble] will have?

4 Outcomes

Three outcomes of interest – authority, pliability, and backlash – are collected directly on the screen following the experimental vignette.³ I detail them below, as well as how I plan to code them based on responses.

1. Authority.

- Question:
 - Imagine there are some important decisions to be made in the village.
- As you probably know, sometimes other people – influential people in

²See Appendix 2 for the Marathi and Hindi versions of the vignette.

³See Appendix 2 for Marathi and Hindi versions of outcome questions.

the village, village elders, other council members, or family members – step in and take village decisions in the place of the president or are generally more active than the president. Sometimes however it is the case that the president is the most influential actor. In this village, do you think that a president like [Bharti/Rohit] [Marathe/Kamble] would be the one who is the most active in taking village decisions, or do you think it would be someone else?

- Response options:
 - President
 - Someone else
- Coding:
 - President = 1, Someone else = 0

2. Pliability (propensity to resign).

- Question:
 - As you probably know, in almost every village, there are generally some influential people when it comes to village affairs, for example a chawri group, or bhauki leaders. Imagine that influential people in this village ask [Bharti/Rohit] [Marathe/Kamble] to resign from their post as president. Do you think that a president like [Bharti/Rohit] [Marathe/Kamble] would accept this request and resign, or would [Bharti/Rohit] [Marathe/Kamble] resist the request?
- Response options:
 - Resign
 - Resist

- Coding:
 - Resign = 1, Resist = 0

3. Backlash.

- Question:
 - Now imagine that [Bharti/Rohit] [Marathe/Kamble] becomes quite powerful as a president. [Bharti/Rohit] [Marathe/Kamble] makes [her/his] own decisions about what is best for the village, which are not always in line with what the influential villagers want, and [she/he] refuses to resign from [her/his] position as president. Do you think that [Bharti/Rohit] [Marathe/Kamble] would likely face backlash from influential people in the village? For example, threats, bullying, or violence?
- Response options:
 - Yes, it is likely
 - No, it is not likely
- Coding:
 - Yes = 1, No = 0

5 Hypotheses and analysis strategy

This section details four main and two secondary hypotheses to be tested, as well as two exploratory analyses. All quantitative analyses will be run via OLS regression, with Huber-White heteroskedasticity-robust standard errors, and statistical significance will be assessed at the 5 % level. I will report covariate-adjusted estimates – my preferred models – in the main paper and unadjusted difference in means estimates in the appendix.

The four main hypotheses and their analysis procedures are as follows:

- M1. The male advantage in authority. I expect that relative to female presidents, hypothetical male presidents will be more likely to be perceived to possess authority (Outcome 1).
 - Estimation: $Authority = \beta_0 + \beta_1 male + \gamma \mathbf{X} + \epsilon$
 - Quantity of interest: β_1 , expected to be positive
 - $\mathbf{X}$: Vector of covariates described in the section below
 - Note: the ATE is a weighted average across the dominant and non-dominant caste treatments.
- M2. The dominant caste advantage in authority. I expect that relative to Dalit presidents, hypothetical Maratha presidents will be more likely to be perceived to possess authority (Outcome 1).
 - Estimation: $Authority = \beta_0 + \beta_1 Maratha + \gamma \mathbf{X} + \epsilon$
 - Quantity of interest: β_1 , expected to be positive
 - $\mathbf{X}$: Vector of covariates described in the section below
 - Note: the ATE is a weighted average across the female and male treatments.
- M3. The male advantage in pliability. I expect that relative to female presidents, hypothetical male presidents will be less likely to be perceived to be willing to resign (Outcome 2).
 - Estimation: $Pliability = \beta_0 + \beta_1 male + \gamma \mathbf{X} + \epsilon$
 - Quantity of interest: β_1 , expected to be negative
 - $\mathbf{X}$: Vector of covariates described in the section below
 - Note: the ATE is a weighted average across the dominant and non-dominant caste treatments.

- M4. The dominant caste advantage in pliability. I expect that relative to Dalit presidents, hypothetical Maratha presidents will likely to be perceived to be willing to resign (Outcome 2).

- Estimation: $Pliability = \beta_0 + \beta_1 Maratha + \gamma X + \epsilon$
- Quantity of interest: β_1 , expected to be negative
- X : Vector of covariates described in the section below
- Note: the ATE is a weighted average across the female and male treatments.

In addition to the four main hypotheses about gender and caste-based authority and pliability differentials, I pre-register two secondary hypotheses and their analysis procedures.

- S1. The dominant caste advantage in backlash. I expect that relative to Dalit presidents, hypothetical Maratha presidents will be less likely to be perceived to face backlash (Outcome 3).

- Estimation: $Backlash = \beta_0 + \beta_1 Maratha + \gamma X + \epsilon$
- Quantity of interest: β_1 , expected to be negative
- X : Vector of covariates described in the section below
- Note: the ATE is a weighted average across the female and male treatments.

- S2. The male advantage in backlash? Relative to female presidents, it is unclear whether hypothetical male presidents will be less likely to be perceived to face backlash (Outcome 3). On one hand, it could be negative, if regardless of community, women who gain power are perceived as a threat to dominance. On the other hand, it could be zero, if women who gain power within the dominant caste-group are not perceived as a threat to dominance. I expect a potential gender effect to be

weaker than the caste effect for this outcome, as political violence in this context often occurs among community lines.

- Estimation: $Backlash = \beta_0 + \beta_1 male + \gamma \mathbf{X} + \epsilon$
- Quantity of interest: β_1 , could be 0 or negative.
- $\mathbf{X}$: Vector of covariates described in the section below
- Note: the ATE is a weighted average across the dominant and non-dominant caste treatments.

Finally, I will perform two exploratory analyses. First, I expect gender dominance and caste dominance to interact, with order of advantage being Maratha men, Maratha women, Dalit men, Dalit women for authority and pliability outcomes. However, this design is underpowered to detect interaction effects, so I will display bar graphs with means for each identity category as a suggestive analysis. Second, I will investigate the qualitative responses to questions 1, 2 and 3, that follow up to ask "why" respondents gave the response that they did.

5.1 Covariates and balance

In the appendix, I will report balance for the following covariates: locality (urban vs. rural), age, gender, religion, caste, education, prior voter participation, self-reported knowledge on local politics, and a gender norms indicator. These are the covariates that I will include in my regression models.

The covariates will be coded as follows:

- Locality (rural area): Factor variable with two levels: Village and Nagar palika or Mahanagar palika.
- Age: continuous variable.

- Gender: Factor variable with two levels: Male, female.
- Religion: Factor variable with seven levels: Hindu, Muslim, Christian, Buddhist, Sikh, Adviasis, and Jain. This is subject to adjustment based on the sample demographics.
- Caste: Factor variable with six levels: Adivasi, Dalit, OBC, Maratha, VJNT, and other general caste. This is subject to adjustment based on the sample demographics.
- Education: Factor variable with three levels: Primary school or less, secondary school, undergraduate and above. This is subject to adjustment based on the sample demographics.
- Prior voter participation: Factor variable with two levels: voted in 2019 state elections or did not vote.
- Self-reported knowledge on local politics: Factor variable with two levels: Not knowledgeable at all or reports having some knowledge.
- Gender norms: The value given for the following question:
 - As per your understanding, approximately what percent (on a scale of 0 to 100) of the inhabitants in your locality think that it is appropriate that women vote? (on a scale of 0 to 100)

In the appendix, I will document analyses that probe balance. I will present the proportion of voters and standard deviation for each covariate level across the four treatment groups. I will assess balance using F-tests of global significance for these covariates for each treatment group, via regressing each of the four treatment group indicators on these covariates, extracting the F-statistic, and its p-value. Failing to reject the null at the 5 percent level supports balance.

5.2 Power calculations

Sample size was fixed and not under control of the researcher. Care was thus taken to increase the intensity of the treatments in order to better chances at higher levels of power. In addition, the preferred specifications include covariate adjustment to try to increase power as blocking was not feasible in the study design. With a sample size of 2,340 voters, I should be able to detect effects for main hypotheses (M1 and M2) of 0.125 of a standard deviation of the outcome, setting alpha at 0.05 (power= 0.80).⁴

References

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⁴EGAP power calculator was used for these estimates.

Appendix 1: Voter Survey Targeted Stratification

Constituency Name	Reserved Status	Total Respondents	Male	Female	Semi-urban/Urban	Rural	Muslim	Dalit (SC)	Adivasi (ST)	OBC and GENERAL
Akkalkuwa	ST	100	50	50	40	60	20	0	60	20
Shahada	ST	100	50	50	40	60	20	20	40	20
Nandurbar	ST	100	50	50	40	60	20	0	40	40
Nawapur	ST	100	50	50	40	60	0	20	60	20
Sakri	ST	120	60	60	40	80	20	20	40	40
Baglan	ST	120	60	60	20	100	0	20	60	40
Shirpur	ST	100	50	50	10	90	0	20	50	30
Sindkheda	General	100	50	50	20	80	0	20	0	80
Dhule Rural	General	100	50	50	0	100	0	20	0	80
Dhule City	General	200	100	100	200	0	100	20	0	80
Malegaon Central	General	220	110	110	220	0	120	20	0	80
Malegaon Outer	General	120	60	60	0	120	40	20	0	60
Chopda	ST	100	50	50	20	80	20	20	40	20
Amalner	General	120	60	60	60	60	20	20	0	80
Jalgaon Rural	General	100	50	50	0	100	0	20	0	80
Jalgaon City	General	220	110	110	220	0	100	20	20	80
Raver	General	120	60	60	40	80	40	20	40	20
Bhusawal	SC	200	100	100	80	120	20	120	20	40
Total		2340	1170	1170	1090	1250	540	420	470	910

Appendix 2: Marathi and Hindi Translations

The survey was carried out in Marathi or Hindi, depending on the preferences of the voter. North Maharashtra is a region with some linguistic heterogeneity as it borders non-Marathi speaking states and also has a non-negligible indigenous population, though the majority of residents in the region still speak and prefer Marathi, the official language of Maharashtra.

Vignette: Marathi

आता मी तुम्हाला आपल्या महाराष्ट्रातल्या गावातील सरपंचाबद्दल सांगणार आहे. मला या सरपंचाबद्दल तुमचे मत जाणून घेण्यात रस आहे. [भारती/रोहित] [मराठे/कांबळे] [त्या/ते] एक 42 वर्षीय [महिला/पुरुष] सरपंच आहेत. [त्या/ते] [मराठा/दलित] सरपंच आहेत. अश्या एका गावाचे सरपंच आहे जिथे अर्ध्यापेक्षा जास्त लोक मराठा आहेत. [त्या/ते] मागील एक वर्षापासून सरपंच आहेत.

ग्राम पंचायत कशी काम करते याची माहिती आणि तुमचा आजपर्यंतचा अनुभव यावरून मला सांगा [भारती/रोहित] [मराठे/कांबळे] ह्यांचा एक सरपंच म्हणून काय अनुभव असेल?

Vignette: Hindi

अब मैं आपको हमारे महाराष्ट्र के गांव के सरपंच के बारे में बताने वाला हूँ। [भारती/रोहित] [मराठे/कांबळे] ये एक 42 वर्षीय [महिला/पुरुष] सरपंच हैं। वह [मराठा/दलित] सरपंच हैं। वह ऐसे एक गांव के सरपंच हैं जहां आधे से ज्यादा लोग मराठा हैं। वह पिछले एक वर्ष से सरपंच हैं।

ग्राम पंचायत कैसे काम करती है इसकी जानकारी और आपके अब तक के अनुभव के आधार पर, बताएं कि एक सरपंच के रूप में [मराठे/कांबळे] का अनुभव कैसा होगा?

Outcomes: Marathi

1. आता कल्पना करा गावात काही महत्वाचे निर्णय घेतल्या जाणार आहेत. जसा तुम्हाला माहिती असेल, कधी कधी इतर गावातील लोक जसे कि - प्रभावी लोक, जेष्ठ मंडळी, इतर सभासद किंवा कुटुंबातील सदस्य - पुढे येतात आणि सरपंचाला डावलून निर्णय घेतात किंवा सरपंचापेक्षा तेच जास्त सक्रिय असतात. परंतु कधी कधी सरपंच सगळ्यात जास्त प्रभावी आहेत. अश्या एखाद्या गावात तुम्हाला काय वाटतं, [भारती/रोहित] [मराठे/कांबळे] निर्णय घेऊ शकतील कि त्यांच्या ऐवजी दुसरेच कोणी निर्णय घेतील?
 - सरपंच
 - इतर कोणीतरी
2. जसा तुम्हाला माहिती असेल, जवळपास प्रत्येक गावात, गावातला कारभार बघायला अनेक प्रभावी लोक असतात जसे चावडीवर बसणारी मंडळी किंवा भाऊबंदकी मधील लोक. कल्पना करा, अश्याच काही मंडळींनी [भारती/रोहित] [मराठे/कांबळे] सरपंच पदाचा राजीनामा द्यायला सांगितले. तुम्हाला काय वाटतं, [भारती/रोहित] [मराठे/कांबळे] सरपंच पदाचा, विनंती स्वीकारून, राजीनामा देतील कि त्याला विरोध करतील?
 - राजीनामा देतील
 - ते विरोध करतील
3. आता कल्पना करा, [भारती/रोहित] [मराठे/कांबळे] एक प्रभावशाली सरपंच [झाल्या/झाले]. [त्या/ते] गावासाठीचे योग्य निर्णय घेतात जे गावातील प्रभावी लोकांच्या नेहमीच मर्जीतले असेलच असा नाही आणि त्या राजीनामा देण्यास नकार देतात. तुम्हाला वाटतं का, [भारती/रोहित] [मराठे/कांबळे] ला गावातील प्रभावशाली व्यक्तीं (जसे - चावडीवर बसणारी मंडळी किंवा भाऊबंदकी मधील लोक) विरोध करतील? उदाहरणार्थ, धमक्या, छळ, किंवा हिंसा?
 - हो, शक्यता आहे
 - नाही, शक्यता नाही

Outcomes: Hindi

1. अब कल्पना कीजिए कि गांव में कुछ महत्वपूर्ण निर्णय लिए जाने वाले हैं। जैसा कि आपको पता होगा, कभी-कभी अन्य गांव के लोग जैसे - प्रभावशाली लोग, वरिष्ठ नागरिक, अन्य सदस्य या परिवार के सदस्य - आगे आते हैं और सरपंच को नजरअंदाज करके निर्णय लेते हैं या सरपंच से अधिक सक्रिय होते हैं। लेकिन कभी-कभी, सरपंच सबसे प्रभावशाली व्यक्ति होते हैं। ऐसे किसी गांव में, आपको क्या लगता है [भारती/रोहित] [मराठे/कांबळे] निर्णय ले पाएंगे / पाएंगी या उनके बजाय कोई और निर्णय लेगा?
 - सरपंच
 - कोई और
2. जैसा कि आपको पता होगा, लगभग हर गांव में, गांव के कामकाज को देखने के लिए कई प्रभावशाली लोग होते हैं, जैसे चावड़ी पर बैठने वाले लोग या भाऊबंदकी के सदस्य। कल्पना करें, ऐसे ही कुछ लोगों ने [भारती/रोहित] [मराठे/कांबळे] से सरपंच पद से इस्तीफा देने के लिए कहा। आपको क्या लगता है, [भारती/रोहित] [मराठे/कांबळे] सरपंच पद से इस्तीफा देने की विनती स्वीकार करेंगे / करेंगी या उसका विरोध करेंगे / करेंगी ?
 - इस्तीफा दे देंगे
 - विरोध करेंगे
3. अब कल्पना कीजिए [भारती/रोहित] [मराठे/कांबळे] एक प्रभावशाली सरपंच हैं। वह गांव के लिए सही निर्णय लेते/ लेती हैं, जो हमेशा गांव के प्रभावशाली लोगों की मर्जी के अनुसार नहीं होते, और वे इस्तीफा देने से इंकार करते/करती हैं। क्या आपको लगता है कि [भारती/रोहित] [मराठे/कांबळे] उनको गांव के प्रभावशाली व्यक्तियों (जैसे - चावड़ी पर बैठने वाले लोग या भाऊबंदकी के सदस्य) का विरोध सहन करना पड़ेगा? उदाहरण के लिए, धमकियां, उत्पीड़न, या हिंसा?
 - हां संभावना है
 - नहीं, संभावना नहीं है